

Yaoyu He

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Education

ShanghaiTech University, Shanghai, China , Computer Science	2022.09 - 2026.07 (Expected)
• GPA: 3.52/4.0 • major GPA: 3.79/4.0 • Coursework: Computer Architecture(GPA: 4.0), Machine Learning(GPA: 4.0), Deep Learning(GPA: 4.0), Artificial Intelligence(GPA: 4.0), Computer Vision(GPA: 4.0)	

Research and Work Experience

LumiAni Technology Co., Ltd Developer and Researcher	2023.06 - 2024.07
• Developed and optimized stable diffusion plugin for 3D reconstruction pipeline • Apply Colmap for camera calibration and 3D Gaussian Splatting for object point cloud reconstruction, and compared with NeRF/NeuS2 • Design LLM-based prompt engineering strategies to automate labeling and analysis of rodent behavior data	
Shanghai Elan Smart Sense Information Technology Co.,LTD.	2024.07 - 2025.01
• Independently responsible for the construction and debugging of LLM modules, using Qwen, flan-t5, gpt2 and other large language models for debugging. Implemented a conversion module that converts IMU semantic information into LLM semantic space. • Use LoRA to fine-tune and optimize the large language model, and use DPO human preferences to optimize the output for the large language model • Processing 30,000+ data entries by aligning text, SMPL motion, and IMU data while performing IMU data cleaning • Lead the website construction part, independently design the website architecture, and lead other students to complete the website construction	
Shanghai Jiaotong RL² lab	2025.02 - 2025.08
• Integrating the G1 humanoid robot into the Robosuite framework with a three-fingered gripper implementation • Implement PPO with curiosity-driven exploration for robot skill acquisition, enabling autonomous adaptation to unseen physical environments through intrinsic reward mechanisms aligned with embodied intelligence principles • Investigate VLA for semantic-aware robotic grasping, leveraging multimodal reasoning to interpret open-vocabulary commands and improve task generalization in real-world scenarios	
ShanghaiTech 4DV Lab	2025.08 - Now
• Work on reinforcement learning with piper robot arm, aim to rotation basketball • Deploy and fine-tuning $pi_{0.5}$ model on libero robot	

Publications

Sophia-in-Audition: Virtual Production with a Robot Performer	2024.02
Taotao Zhou, Teng Xu, Dong Zhang, Yuyang Jiao, Peijun Xu, Yaoyu He , Lan Xu, Jingyi Yu <i>Proceeding of the 32nd ACM International Conference on Multimedia</i> 2402.06978	
Mojito: LLM-Aided Motion Instructor with Jitter-Reduced Inertial Tokens	2025.02
Ziwei Shan, Yaoyu He , Chengfeng Zhao, Jiashen Du, Jingyan Zhang, Qixuan Zhang, Jingyi Yu, Lan Xu <i>arxiv preprint</i> 2502.16175	
From Noise to Intent: Anchoring Generative VLA Policies with Residual Bridges	2026.01

Yiming Zhong, **Yaoyu He**, Zemin Yang, Pengfei Tian, Yifan Huang, Qingqiu Huang, Xinge Zhu, Yuexin Ma

Forty-Third International Conference on Machine Learning
2604.21391 [🔗](#)

Projects

I am VQ-VAE

[I am VQ-VAE 🔗](#)

- Using VQ-VAE-based tokenizer, mitigating noise and drift
- Using Transformer-based generator, generate motion conditioned on language instruction

Chinese Checkers AI Agent

[Chinese Checkers 🔗](#)

- Developed hybrid RL agent combining Approximate Q-learning with Monte Carlo Tree Search
- Implemented neural heuristic evaluator using self-play reinforcement learning

Coursebench

[coursebench-frontend 🔗](#)

- Develop a website about the course selection page

Technologies

Languages: Python, C++, C, JavaScript, Typescript, Rust, Java, C#,

Technologies: PyTorch, Mujoco, Gym, Qt, Unity, React, Vue, SpringBoot, Tauri